

Cucineanu Dragos Adrian

grupa 231

i = 9

1. multimea  $(0, 9] \cap \mathbb{N} = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$

cardinalul multimei = 9 numere

a) fie  $x$  un număr din multime

$x : 3 \Rightarrow x \in \{3, 6, 9\}$

~~Ră~~  $p = \frac{\text{nr cazuri favorabile}}{\text{nr cazuri posibile}} = \frac{3}{9} = \frac{1}{3} = 0,3333$

b) ~~fi~~ fie  $y$  un număr din multime

$y$  nr pătrat perfect  $\Rightarrow y \in \{1, 4, 9\}$

$p = \frac{\text{nr cazuri favorabile}}{\text{nr cazuri posibile}} = \frac{3}{9} = \frac{1}{3} = 0,3333$

c) fie  $z$  un număr din multime

$z$  nr prim  $\Rightarrow z \in \{2, 3, 5, 7\}$

$p = \frac{\text{nr cazuri favorabile}}{\text{nr cazuri posibile}} = \frac{4}{9} = 0,4444$

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$$2. \quad X \sim \begin{pmatrix} -1 & 0 & 1 \\ \frac{9}{1000} & \frac{1}{100} & x \end{pmatrix}$$

$$a) \quad \frac{9}{1000} + \frac{1}{100} + x = 1$$

$$x = 1 - \frac{1}{100} - \frac{9}{1000}$$

$$= 1 - \frac{19}{1000} = \frac{981}{1000} = 0,981$$

$$b) \quad \text{Var}(X) = E(X^2) - E(X)^2$$

$$E(X) = (-1) \cdot \frac{9}{1000} + 0 \cdot \frac{1}{100} + 1 \cdot \frac{981}{1000}$$

$$= -\frac{9}{1000} + \frac{981}{1000} = \frac{972}{1000} = 0,972$$

$$E(X^2) = (-1)^2 \cdot \frac{9}{1000} + 0^2 \cdot \frac{1}{100} + 1^2 \cdot \frac{981}{1000}$$

$$= \frac{9}{1000} + \frac{981}{1000} = \frac{990}{1000} = 0,99$$

$$\text{Var}(X) = E(X^2) - E(X)^2 = 0,99 - (0,972)^2$$

$$= 0,99 - (0,944784) = 0,99 - 0,944784$$

$$= 0,045216$$

$$= 0,0452$$

$$1) \text{Var}(2x) = 2^2 \text{Var}(x)$$

$$= 4 \text{Var}(x)$$

$$= 4 \cdot 0,0277$$

$$= 0,1108$$

$$= 4 \cdot 0,0453$$

$$= 0,1812$$

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3. domeniul de valori  $[0, 9]$

$$f(x) = c \cdot x \cdot y$$

$$a) \int_0^y \int_0^y c \cdot x \cdot y \, dx \, dy = 1$$

$$\int_0^y c \cdot y \cdot \frac{x^2}{2} \Big|_0^y \, dy =$$

$$\int_0^y c \cdot y \cdot \frac{(y^2 - 0)}{2} \, dy =$$

$$\int_0^y \frac{c \cdot y^3}{2} \, dy =$$

$$\frac{c \cdot y^4}{2} \Big|_0^y =$$

$$\frac{c \cdot y^4}{2} \Big|_0^y = \frac{(y^4 - 0)}{2} =$$

$$\left(\frac{y^4}{2}\right) \Big|_0^y = \frac{y^4 - 0}{2} = 1 \Rightarrow c = \frac{4}{y^4}$$

$$b) P(x \leq 3) = \int_0^y \int_0^3 c \cdot x \cdot y \, dx \, dy =$$

$$= \int_0^y c \cdot y \cdot \frac{x^2}{2} \Big|_0^3 \, dy$$

$$= \int_0^y \frac{c \cdot y^3}{2} \, dy$$

$$= \frac{c \cdot y^4}{2} \Big|_0^y$$

$$= \frac{c \cdot y^4}{2} \Big|_0^y = \frac{y^4}{2} \cdot \frac{4}{y^4} = \frac{1}{2}$$

c)  $x, y$  independiente  $\Leftrightarrow f(x) = f_x(x) \cdot f_y(y)$

$$f_x(x) = \int_0^y f(x) dy = \int_0^y cxy dy = c x \frac{y^2}{2} \Big|_0^y = \frac{y^2}{2} c x$$

$$f_y(y) = \int_0^y f(x) dx = \int_0^y cxy dx = c y \frac{x^2}{2} \Big|_0^y = \frac{y^2}{2} c y$$

$$f_x(x) \cdot f_y(y) = \frac{y^2}{2} c x \cdot \frac{y^2}{2} c y = \frac{y^4}{4} c^2 x y =$$

$$= \frac{6561}{4} \cdot \frac{4}{6561} \cdot cxy = cxy = f(x) \Rightarrow$$

$$f_x(x) \cdot f_y(y) = f(x) \Rightarrow x, y \text{ independiente}$$

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$$4. P(D_H | A) = 0,5$$

$$P(A) = \frac{9}{200} = 0,045$$

$$P(D_H | B) = 0,9$$

$$P(B) = \frac{200 - 9}{200} = \frac{191}{200} = 0,955$$

Probabilitatea predictivă a priari:

$$\begin{aligned} P(D_H) &= P(D_H | A) \cdot P(A) + P(D_H | B) \cdot P(B) \\ &= 0,5 \cdot 0,045 + 0,9 \cdot 0,955 \\ &= 0,0225 + 0,764 = 0,7865 \end{aligned}$$

Probabilitatea predictivă a posteriori:

Label Bayes:

| H     | P(H)  | P(D   H) | P(D   H) · P(H) | P(H   D)                         |
|-------|-------|----------|-----------------|----------------------------------|
| A     | 0,045 | 0,5      | 0,0225          | $\frac{0,0225}{0,7865} = 0,0286$ |
| B     | 0,955 | 0,9      | 0,764           | $\frac{0,764}{0,7865} = 0,9713$  |
| total | 1     |          | 0,7865          | 1                                |

$$\begin{aligned} P(D_H | D) &= P(D_H | A) \cdot P(A | D) + P(D_H | B) \cdot P(B | D) \\ &= 0,5 \cdot 0,0286 + 0,9 \cdot 0,9713 \\ &= 0,0143 + 0,777 \\ &= 0,7913 \end{aligned}$$

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$$5. \quad 110 \text{ anunțuri} \begin{cases} 9 \text{ averuiri} \\ 101 \text{ nereveruiri} \end{cases}$$

Decarele datele provin dintr-o repartiție binomială

$$(110, \theta), \text{ verosimilitatea este } p(X_1 | \theta) = C_{110}^9 \theta^9 (1-\theta)^{101}$$

Tabelul de actualizare Bayesiană:

| ipoteză  | a priori    | verosimilitate                        | numărător Bayes                                            |
|----------|-------------|---------------------------------------|------------------------------------------------------------|
| $\theta$ | $1 d\theta$ | $C_{110}^9 \theta^9 (1-\theta)^{101}$ | $C_{110}^9 \theta^9 (1-\theta)^{101} d\theta$              |
| total    | 1           |                                       | $T = C_{110}^9 \int_0^1 \theta^9 (1-\theta)^{101} d\theta$ |

a posteriori

$$c \theta^9 (1-\theta)^{101}$$

1

Observația are loc cu  $a = 10$  și  $b = 102$ , de aceea  
pdf a posteriori

$$f(\theta, X_1) = c \theta^9 (1-\theta)^{101}$$

are o repartiție beta  $(10, 102)$  și constanta de norma-  
lizare  $c$ :

$$c = \frac{111!}{9! \cdot 101!}$$

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6. Intervalul simetric de  $\frac{9}{100}$  probabilitate pentru normala standard.

$$\text{fie } p = \frac{9}{100} = 0,045$$

~~Intervalul~~ intervalul  $[z_1, z_t]$

intervalul  $p$ -probabilitate este intervalul  $[z_1, z_t]$ ,

$$\text{cu } t - z = p$$

$$P(0 \leq X \leq t) = 0,045$$

$$P(0 \leq X \leq 0) = P(t \leq X \leq 1) = \frac{1 - 0,045}{2} = 0,4775$$

$$\text{intervalul} = [20,4775, 20,5225]$$

Urmind R calculăm:

$$20,4775 = \Phi_{\text{norm}}(0,4775; 0, 1) = -0,0564$$

$$20,5225 = \Phi_{\text{norm}}(0,5225, 0, 1) = 0,0564$$

$$\text{interval} = [-0,0564, 0,0564]$$